Learning to Play Risk with Action-Injected Graph Neural Networks

A graph-attention reinforcement-learning approach, compared across DQN, Dueling DQN, and PPO

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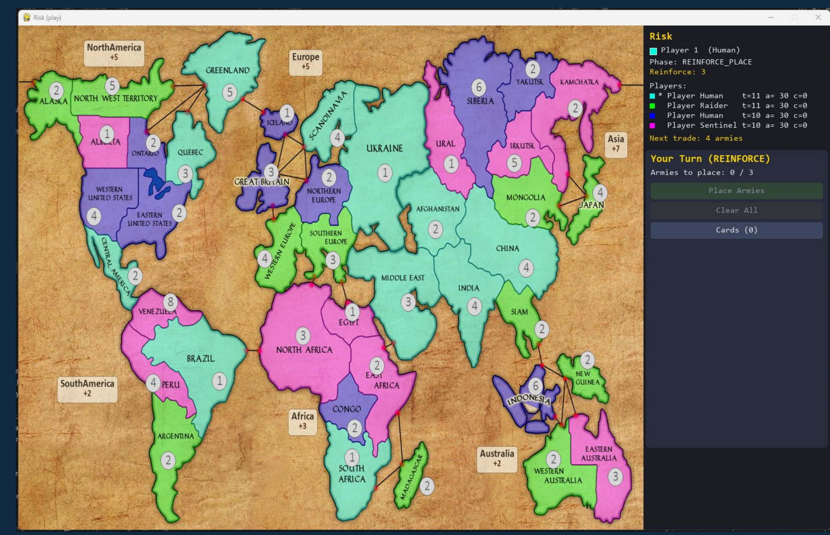

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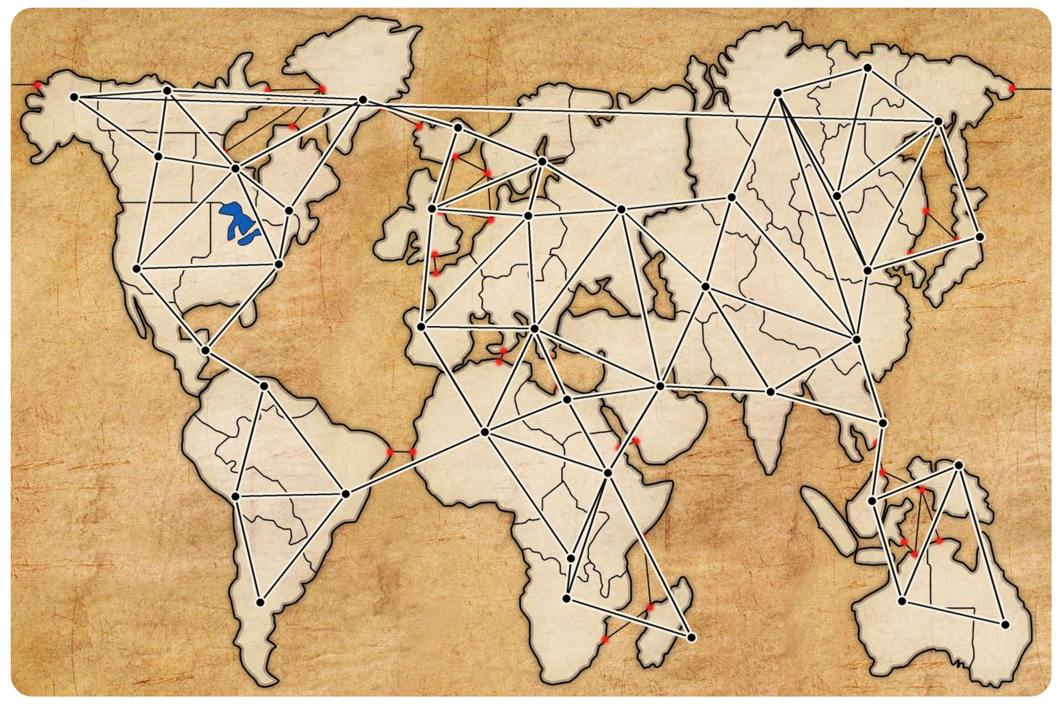
PROJECT REPOSITORY
GitHub

Instead of scoring one enormous fixed move list, the system injects each legal Risk move into the board graph and scores that state–action graph with one shared graph-attention encoder.

1 Board as a graph

Risk's territories and borders are relational: move quality depends on neighbours, armies, and contested continents.

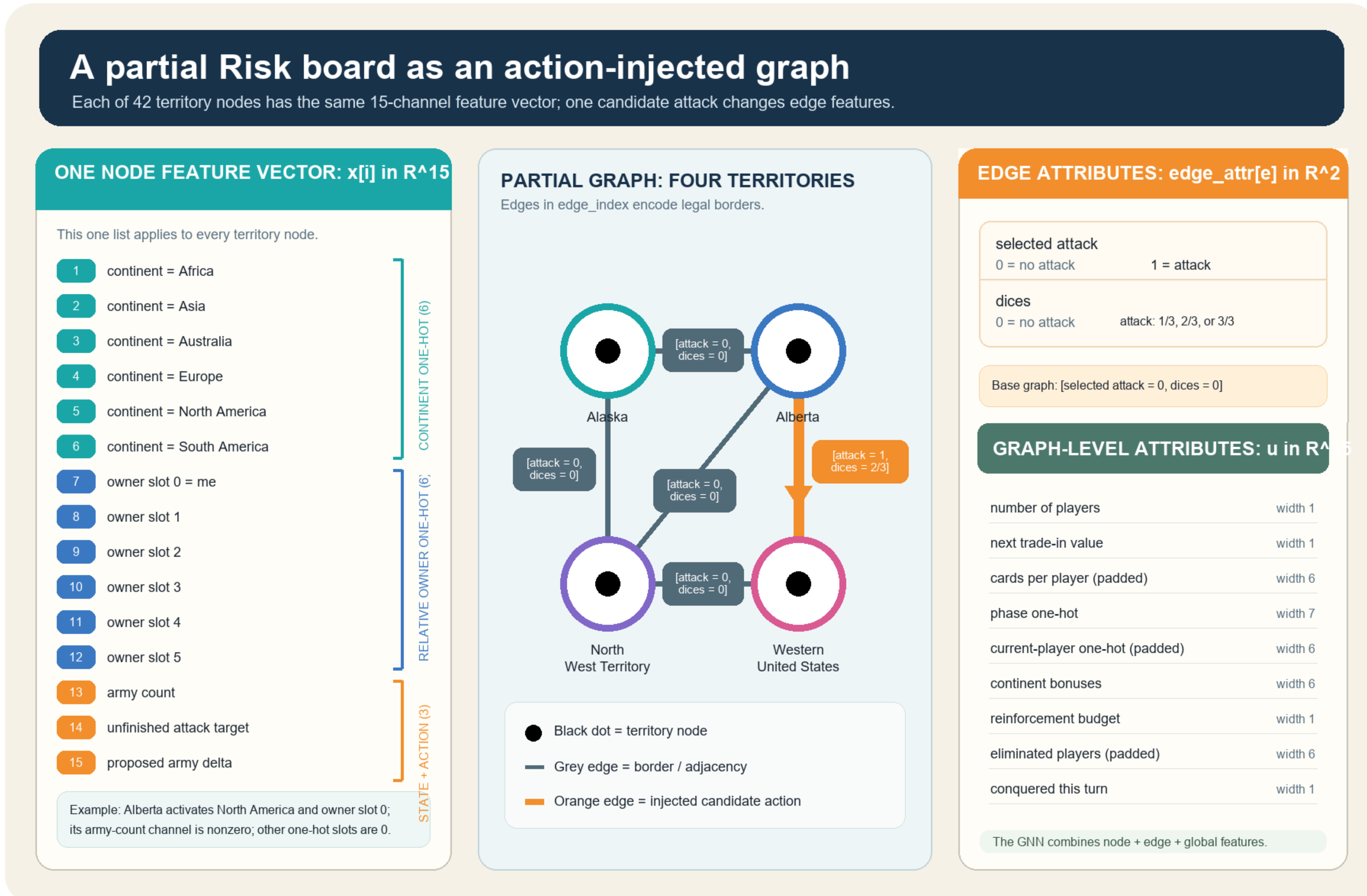


42 territory nodes | 83 borders | 166 directed edges

X [42 x 15] E [166 x 2] u [1 x 35]

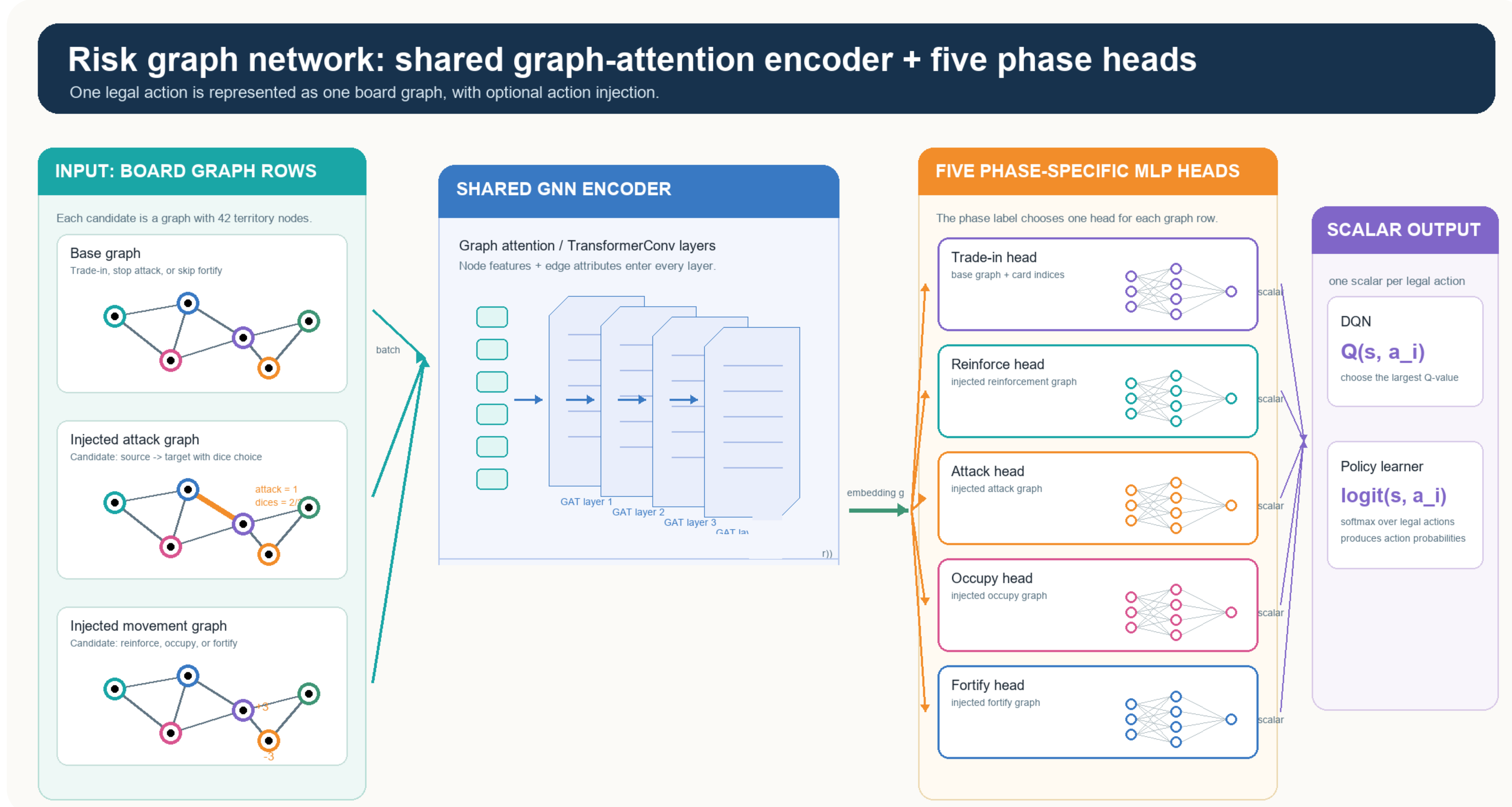
2 Inject each legal candidate action

The action becomes part of the graph: an attack changes one border; other phases mark affected territory rows.



3 Shared GATN encoder + phase heads

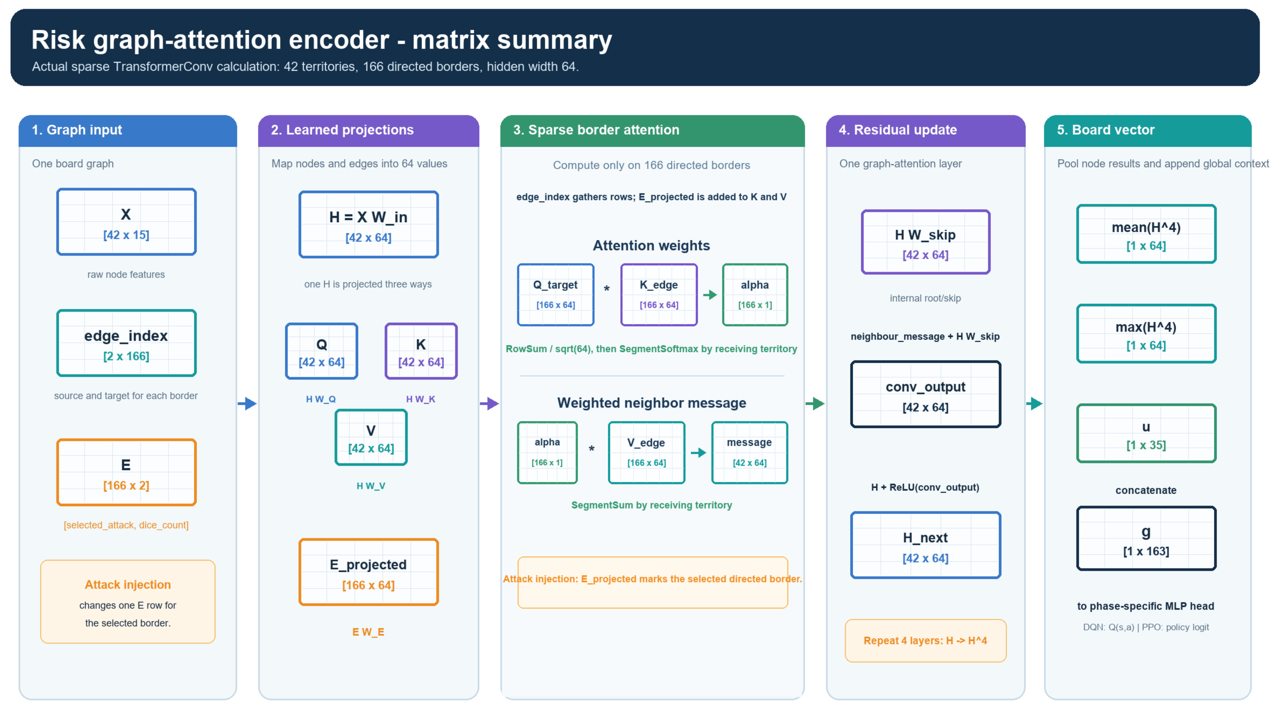
Four TransformerConv layers read the injected graph; the current Risk phase selects its small scoring head.



How GATN attention uses the injected action

GATN attends only across neighboring territories. Q represents the receiving territory; K determines how relevant each neighbor is; V is the information that neighbor sends. The injected action changes K and V on the affected border, changing its attention weight and message.

Four graph-attention layers aggregate these weighted neighbor messages—never all territory pairs.



Reward shaping

Reward = terminal (+100 / -100) + 0.1 x local action terms + 0.1 x board progress after opponents.

Local shaping applies only to learner decisions; board progress is separately scaled (not clipped) after every opponent has played.

Phase / trigger	Key DQN_103 calculation	What it teaches
Terminal (game ends)	+100 win; -100 loss (unscaled)	Winning stays the main objective.
Trade-in (optional at 3-4 cards)	+/- 0.30 x hand factor / card-set value +0.60 if a traded card matches an owned territory	Save an optional trade unless it creates value.
Reinforce (place armies)	+1.2 x share placed on a territory bordering an enemy +1.5 x [min(own / weakest adjacent enemy, 2.5) - 1] -0.8 on a safe interior territory	Build an attack-capable border; do not park armies where they cannot fight.
Attack (choose / resolve)	2 x [min(attacker / defender, 3) - 1.5] +0.6 x (defender losses - attacker losses) +1.2 conquest; +4 continent; +4 + 1.5/card elimination	Use strong odds, maximum dice, and profitable battles.
Occupy + fortify (move armies)	+ moved / movable into a captured territory or toward an enemy border Penalty for moving away or leaving two borders badly balanced	Put force where an opponent can attack, then keep it there.
Board progress (after opponents)	20 x territory-share change; 0.1 x army-share change 2.5 x continent-bonus change; extra continent-loss penalty	Reward gains that survive a full opponent round.

Results: training + held-out evaluation

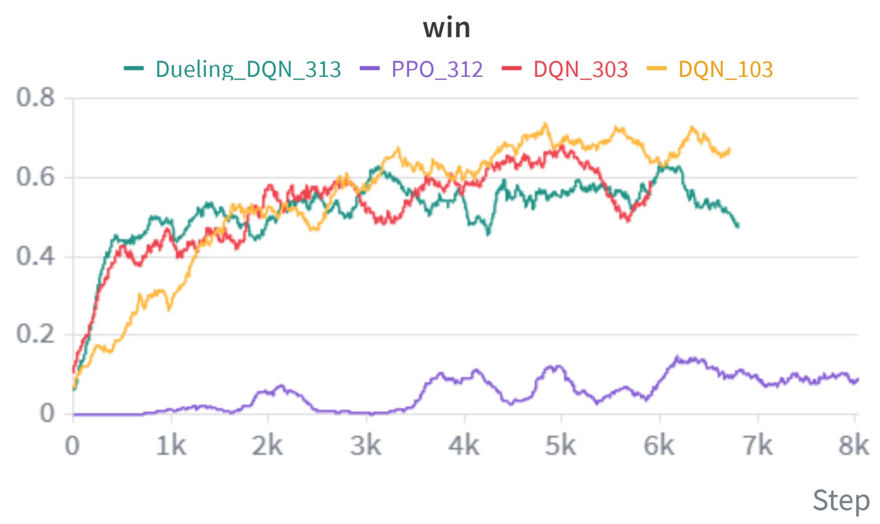
DQN learning; learner comparison; DQN evaluated across 3–6 players.

Win Rate

DQN | Dueling DQN | PPO

DQN and **Dueling DQN** improve quickly and reach similar win rates.

PPO improves only slightly and stays far behind, showing poor fit for this game.

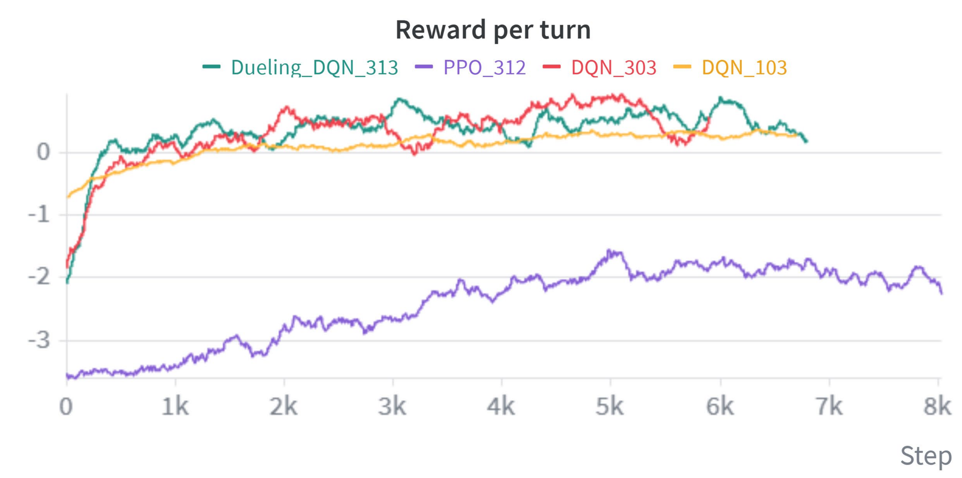


Learner comparison

DQN | Dueling DQN | PPO

DQN and **Dueling DQN** move from negative rewards to stable positive rewards per turn.

PPO also improves but remains strongly negative throughout training.



DQN evaluation

Top 5 checkpoints · 54 games each

54 games / checkpoint
3 seeds · every learner seat
ε = 0 · 2,000-step cap

Latest ep006700
46 / 54 wins = 85.2%
0 timeouts